

Seohyun Park

(770) 688-7065 | seopark935@gmail.com | <http://www.linkedin.com/in/spark935> | seopark935.github.io | US Citizen

Education

Georgia Institute of Technology | Atlanta, GA, USA

August 2024 – Present

Bachelor of Science in Computer Engineering, GPA 4.0/4.0

Expected Graduation: May 2027

- *Relevant Coursework:* Circuit Analysis, Programming for HW/SW Systems, Data Structures and Algorithms, Architecture, Systems, Concurrency and Energy in Computation, VLSI and Advanced Digital Design

Skills

EDA & Digital Design: Cadence Virtuoso (RTL-to-GDSII Certified), RTL (VHDL/SystemVerilog), Quartus Prime

Programming & Software: C/C++, Python, Java, RISC-V Assembly, MATLAB, Git, Linux, Atlassian Jira

Hardware & Tools: Altera FPGA, oscilloscope, logic analyzer, multimeter

Experience & Activities

Liang Lab | Emory Department of Biochemistry, Atlanta

August 2022 – Present

Computational Researcher

- Engineered a high-performance distributed pipeline using OpenMPI and Python to parallelize docking of 1.9M+ molecules across a 192-core cluster, reducing runtime from weeks to days.
- Automated data preprocessing and iterative filtering with OpenBabel, improving throughput and reproducibility of large-scale screening workflows.

SiliconJackets | Georgia Tech, Atlanta

August 2024 – Present

Physical Design Engineer

- Designed and optimized fetch block microarchitecture within a 6-stage RISC-V core, reducing area inefficiencies and improving estimated frequency by ~5%.
- Worked within full RTL-to-GDSII flow (Cadence synthesis, place & route, signoff) on SkyWater 130nm process, integrating design into Caravel SoC harness.

EcoCar VIP (Vertically Integrated Project) | Georgia Tech, Atlanta

August 2025 – Present

Controlled and Automated Vehicles (CAVs) Engineer

- Optimized Python perception modules in autonomous driving stack, improving sensor-fusion accuracy and object detection reliability; built simulation framework to validate lane-tracking algorithms pre-deployment.
- Processed raw vehicle .blf log files into structured datasets for visualization and simulation comparison; built Python scripts to scrape signals and compute derived metrics (acceleration, jerk, etc.), streamlining analysis of recorded vs. simulated performance.

URS Compliance Services | Duluth, Georgia

June – July 2022

Data Analysis Intern

- Verified and managed 500+ legal entity records in state/internal systems.
- Built Excel/Python macros to streamline contract processing for 300+ documents.

Projects

5-Stage Pipelined RISC-V Processor | Georgia Tech, ECE 3058

May 2025 – July 2025

- Implemented a 5-stage pipelined RV32I processor in SystemVerilog with hazard detection, stalling, data forwarding, and pipeline flushing; validated using ModelSim and benchmark programs.
- Built on a prior single-cycle RV32I design supporting ADD, ADDI, SUB, LW, SW, SLT, and JAL, with integrated instruction/data memory.

FPGA Morse Code LED Controller | Georgia Tech, ECE 2031

January 2025 – May 2025

- Programmed a VHDL-based LED peripheral on the DE10 FPGA that converts ASCII input into Morse code patterns and displays them via LEDs and a 7-segment display.
- Implemented static and flashing modes via state machine design to interface with an SCOMP processor without altering internal architecture.

Snake Game on Mbed Platform | Georgia Tech, ECE 2035

January 2025 – May 2025

- Developed an interactive Snake game in C++, integrating hardware elements (Mbed microcontroller, nav-switch input, uLCD display), with game features including score tracking, win/lose states, and real-time collision detection.